



Cambridge Lower Secondary Checkpoint

SCIENCE

0893/02

Paper 2

April 2024

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Markers were instructed to award marks. It does not indicate the details of the discussions that took place at a Markers' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the End of Series Report. Cambridge will not enter into discussions about these mark schemes.

This document has **14** pages. Any blank pages are indicated.

General guidelines on marking

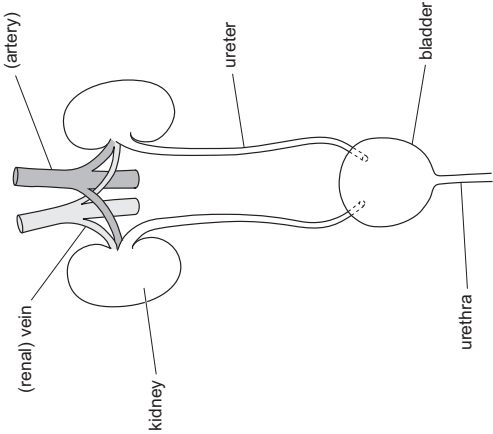
Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed so answers do not need to be grammatically correct. Significant figures will be indicated in the question or in the mark scheme. Unless specified all marking points are independent.

Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>Underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working, provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

Question	Answer	Marks	Further Information
1(a)		3	all five correct = 3 marks three or four correct = 2 marks one or two correct = 1 mark
1(b)	(idea that) child B has an extra ureter or child B has more ureters or child B has 3 ureters (not 2)	1	Accept ecf for name of ureter from (a) Accept ora for child A Ignore child A has a normal renal system or child B has an abnormal renal system

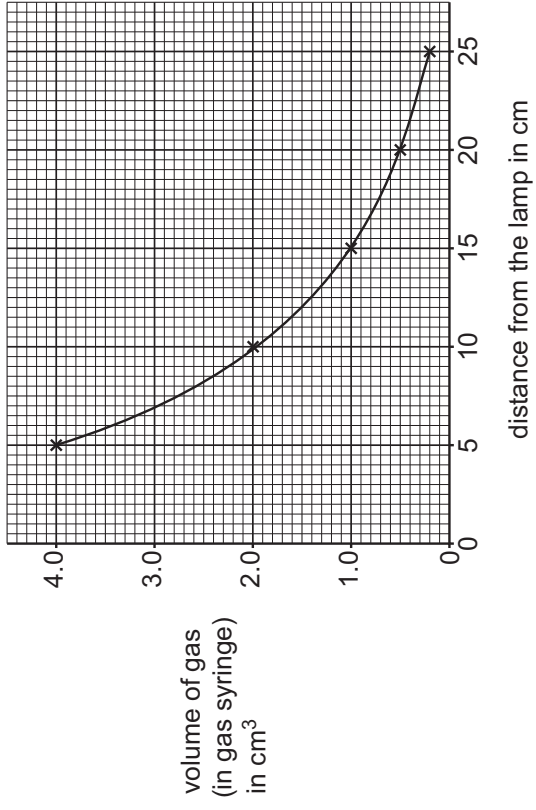
Question	Answer	Marks	Further Information
2(a)	A	1	
2(b)	protons have a positive charge and electrons have a negative charge (idea that) fewer electrons than protons or has three protons and two electrons or has more protons than electrons	2	each correct answer = 1 mark Accept more positive particles than negative particles or fewer negative particles than positive particles
2(c)	A	1	
2(d)	attraction between a positive(ly) (charged) ion and a negative(ly) (charged) ion	1	Accept electrostatic attraction between ions Accept attraction between oppositely charged ions

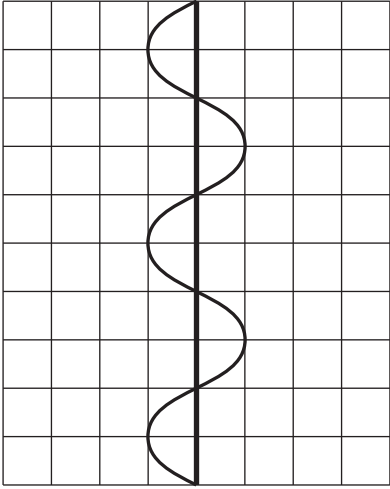
Question	Answer	Marks	Further Information
3(a)	(amplitude) increases or (amplitude) is higher	1	Ignore gets louder
3(b)	frequency	1	Ignore descriptions of the word frequency

Question	Answer	Marks	Further Information
4(a)	convection (currents)	1	
4(b)	they move towards each other or converge or subduct	1	Accept they collide or pushed against each other or they hit each other or crash into each other or move against each other Accept (fold) mountain building or volcanoes or volcanic eruptions or earthquakes

Question	Answer	Marks	Further Information
5	(density =) mass ÷ volume or $840 \div 80$ 10.5 g/cm ³	3	10.5 g/cm ³ with or without working = 3 marks 10.5 with or without working = 2 marks correct unit = 1 mark equation or evidence of equation, e.g. correct substitution = 1 mark Accept $(d =) \frac{m}{v}$

Question	Answer	Marks	Further Information
6(a)	any one from pattern of alignment is the same either side of the ridge rocks of similar age are being created or rocks of similar age are added to plates either side of ridge the sea floor is moving apart	1	Accept (idea that) mid-Atlantic ridge present (in the correct context) or (idea that) new rock or sea floor is formed as the plates move apart or (idea that) magma moves upwards into plate boundary Note tectonic plates move is not sufficient
6(b)	(idea that) similar fossils are found in different continents (that are not connected)	1	Accept (idea that) similar fossils are found in different countries (that are not connected) Accept (idea that) similar fossils found on different tectonic plates Accept same fossils for similar fossils Accept (idea that) similar fossil found in different places or in different parts of the world

Question	Answer	Marks	Further Information
7(a)	oxygen	1	Accept O₂ Ignore O Note if name and formula given the name takes precedence
7(b)		3	labelling and units of x-axis and labelling and units of y-axis = 1 mark Accept distance in cm and volume in cm³ all points correctly plotted to $\pm$ half a small square = 1 mark curve of best fit = 1 mark Accept ecf from incorrect plots Do not accept straight lines between the dots
7(c)	as distance from lamp increases volume of gas collected decreases	1	Accept negative correlation Accept ora Ignore inversely proportional
7(d)	to make chlorophyll	1	

Question	Answer	Marks	Further Information
8(a)	 waveform C	1	
8(b)(i)	(waveform F) _____	1	Accept a horizontal straight line by eye
8(b)(ii)	(sound waveform F) has no sound	1	Accept (loudness) decreases Accept ora if D specified Accept ecf from (b)(i)

Question	Answer	Marks	Further Information
9(a)	environmental change	1	more than one box ticked = 0 Accept any indication of the correct answer, e.g. circling or underlining, but ticking takes precedence
9(b)(i)	mammals	1	
9(b)(ii)	(idea that) coral has a high(er) percentage of low conservation (compared to birds) or birds have more medium conservation	1	Accept coral has less medium conservation Accept ora Accept data used from graph, e.g. high and medium is approximately 55% for birds but 35% for coral

Question	Answer	Marks	Further Information
10	(resistance =) voltage $\div$ current or $2.7 \div 5.0$ (resistance =) $0.54 (\Omega)$	2	0.54 with or without working = 2 marks equation or evidence of equation, e.g. correct substitution = 1 mark Accept $V \div A$ or $V \div I$ or $V \div C$

Question	Answer	Marks	Further Information
11(a)	(description) mass remains the same (explanation) (idea of) law of conservation of mass	2	each correct answer = 1 marks Accept mass cannot be created or destroyed (during a chemical reaction) / no mass is gained or lost (in a chemical reaction) Accept atoms cannot be lost or gained (in a chemical reaction)
11(b)	iron + copper sulfate $\rightarrow$ iron sulfate + copper	1	Accept reactants in either order and products in either order Accept correct formulae, but words take precedence

Question	Answer	Marks	Further Information
12(a)	climate change	1	Accept greenhouse effect or global warming or increasing carbon dioxide in the air or increasing temperature (of Earth) Accept ice caps melt or glaciers melt or sea ice melts or icebergs melt
12(b)	flooding or loss of land	1	Accept loss of habitat Accept some (named) animals or (named) plants become endangered / some (named) animals or (named) plants become extinct Accept human populations living by coast need to move or population on islands will need to move Ignore sea level rise Ignore increase in rain fall or extreme weather Ignore destruction of property Ignore tsunami

Question	Answer	Marks	Further Information
13(a)	caesium or francium	1	Accept Cs or Fr, but name takes precedence.
13(b)	any value between 950 and 1500 (°C)	1	
13(c)	simple (structure)	1	

Question	Answer	Marks	Further Information
14(a)	(lamp added in) series current decreases	2	each correct answer = 1 mark Accept circuit has less current Note resistance decreases and current decreases = 0 marks
14(b)		1	Note a rectangle on its own is not sufficient and the wires at the end should not have any gaps Do not accept a symbol for a variable resistor or with R in rectangle or horizontal line through rectangle

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Question	Answer	Marks	Further Information
15(a)	C E (A) D B	2	C and E correct = 1 mark D and B correct = 1 mark
15(b)	nurseries	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence

Question	Answer	Marks	Further Information
16(a)	(beaker 1) 28.5 (beaker 2) 21.0	2	each correct answer = 1 mark Accept 21 (for beaker 2)
16(b)	(beaker 1) 7.5 and (beaker 2) 1.5	1	Accept ecf
16(c)	dark(er) or black and speed up or increase	1	Accept ora for a light(er) colour

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